

A brief introduction to mixing time in Markov chains

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Plan of the talk

1. Markov chains

- Shuffling a deck of cards: how random is random?
- Definitions, equilibrium, convergence.

2. Mixing time and the cutoff phenomenon

- Distance to equilibrium, mixing time, spectral gap.
- Cutoff: the chain becomes random *abruptly*.

3. The simple exclusion process in dimension one

- k indistinguishable particles on a segment.
- Sharp answer: cutoff at $\frac{N^2 \log k}{2\pi^2}$, and why.

Goal: everything is elementary; the questions are not.

Part 1: Markov chains

A question from everyday life

Question

You shuffle a deck of 52 cards. How many shuffles until the deck is “random”?

To even ask the question we need:

- a **state space**: the $52! \approx 10^{68}$ orderings of the deck;
- a **random rule** to pass from one ordering to the next (one shuffle);
- a notion of “**random**”: the uniform measure on the $52!$ orderings;
- a way to **measure the distance** between the law of the deck after t shuffles and the uniform measure.

The first three items make a Markov chain; the fourth one leads to the mixing time.

Markov chains: discrete time

Ω : a finite set (the **state space**).

Definition

$P = (P(x, y))_{x, y \in \Omega}$ is a **transition matrix** if

$$P(x, y) \geq 0 \quad \text{and} \quad \sum_{y \in \Omega} P(x, y) = 1 \quad \forall x \in \Omega.$$

A random sequence $(X_t)_{t \in \mathbb{N}}$ is a Markov chain with transition matrix P if

$$\mathbb{P}[X_{t+1} = y \mid X_t = x, X_{t-1}, \dots, X_0] = \mathbb{P}[X_{t+1} = y \mid X_t = x] = P(x, y).$$

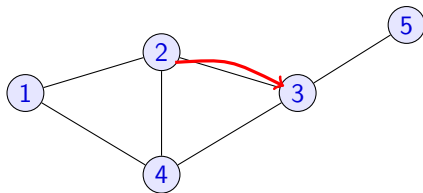
The Markov property: to predict the future, the present is enough — the past may be forgotten.

Iterating: $\mathbb{P}[X_t = y \mid X_0 = x] = P^t(x, y)$, the t -th power of the matrix.

Example 1: random walk on a graph

$G = (V, E)$ a finite connected graph. From x , jump to a uniformly chosen neighbour:

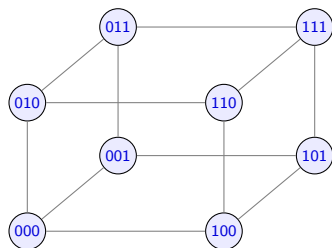
$$P(x, y) = \frac{1}{\deg(x)} \mathbf{1}_{\{x \sim y\}}.$$



- Card shuffling is a random walk on a graph whose vertices are the [52!](#) orderings.
- The [simple exclusion process](#) of Part 3 is a random walk on a graph with $\binom{N}{k}$ vertices.

Example 2: the hypercube (or: n independent coins)

$\Omega = \{0,1\}^n$, $|\Omega| = 2^n$. At each step, pick a coordinate $i \in \llbracket 1, n \rrbracket$ uniformly and **resample** x_i by a fair coin.



Why this example? It is a toy model where **everything** can be computed by hand, and it already displays the phenomenon (cutoff) we are after.

Equilibrium: the uniform measure on $\{0,1\}^n$.

Continuous time: the generator

- **Generator:** $\mathcal{L} = (r(x, y))_{x, y \in \Omega}$, an $\Omega \times \Omega$ matrix with
 - ▶ off-diagonal entries $r(x, y) \geq 0$ (jump rates),
 - ▶ every row sum equal to zero.

For $f : \Omega \rightarrow \mathbb{R}$,

$$(\mathcal{L}f)(x) := \sum_{y \in \Omega} r(x, y) (f(y) - f(x)).$$

- **Markov semi-group** $(P_t)_{t \geq 0}$:

$$P_t := e^{t\mathcal{L}} = \sum_{m \geq 0} \frac{(t\mathcal{L})^m}{m!}, \quad P_t(x, y) \geq 0, \quad \sum_{y \in \Omega} P_t(x, y) = 1.$$

Picture: on each oriented pair (x, y) put an independent Poisson clock of rate $r(x, y)$; when it rings and the chain sits at x , it jumps to y .

Continuous time removes periodicity issues and makes formulas cleaner — we use it from now on.

Equilibrium

- μ is an invariant probability measure if

$$(\forall t \geq 0, \mu P_t = \mu) \Leftrightarrow \mu \mathcal{L} = 0.$$

- Irreducible: for all $x \neq y$ there is a path $x = z_0, z_1, \dots, z_\ell = y$ with $r(z_{m-1}, z_m) > 0$ for all m .

Theorem (Convergence to equilibrium)

If $(\Omega, \mathcal{L})$ is irreducible, there is a unique invariant probability measure μ , and for every initial distribution ν ,

$$\lim_{t \rightarrow \infty} \sum_{y \in \Omega} \left| \mathbb{P}^\nu [X_t = y] - \mu(y) \right| = 0.$$

This is qualitative. The whole talk is about making it quantitative:
how long do we have to wait?

Reversibility

- Detailed balance: for all $x, y \in \Omega$,

$$\mu(x)r(x, y) = \mu(y)r(y, x) \quad \implies \quad \mu\mathcal{L} = 0.$$

(Much easier to check than $\mu\mathcal{L} = 0$: it is a local identity.)

- Then $\mathcal{L}$ is self-adjoint in $\langle f, g \rangle_\mu := \sum_x \mu(x)f(x)g(x)$, and there is an orthonormal basis of eigenfunctions (φ_i) ,

$$\mathcal{L}\varphi_i = -\lambda_i\varphi_i, \quad 0 = \lambda_1 < \lambda_2 \leq \dots \leq \lambda_{|\Omega|}, \quad P_t\varphi_i = e^{-\lambda_i t}\varphi_i.$$

Moral

For a reversible chain, convergence to equilibrium is governed by the spectrum of $\mathcal{L}$ — linear algebra enters the picture.

Example: random walk on a graph is reversible with respect to $\mu(x) \propto \deg(x)$.

Why do we care?

- **Markov Chain Monte Carlo.** To sample from a complicated measure μ (statistical physics, Bayesian statistics, combinatorial counting), build a chain with invariant measure μ and run it.
How long must we run it? $\Rightarrow$ mixing time.
- **Statistical mechanics.** The relaxation of a physical system to equilibrium; phase transitions are often visible in the mixing time.
- **Card shuffling / combinatorics.** Random permutations, random walks on groups.
- **Algorithms.** Approximate counting (permanents, colourings, volumes) reduces to sampling, hence to mixing.

The mixing time is the running time of the algorithm.

Part 2: Mixing time and cutoff

Distance to equilibrium

- **Total variation distance** between two probability measures α, β on Ω :

$$\|\alpha - \beta\|_{\text{TV}} := \sup_{A \subset \Omega} |\alpha(A) - \beta(A)| = \frac{1}{2} \sum_{x \in \Omega} |\alpha(x) - \beta(x)|.$$

(A = a statistical test; $\|\alpha - \beta\|_{\text{TV}}$ = how well the best test distinguishes α from β .)

- **Distance to equilibrium** (worst starting point):

$$d(t) := \max_{x \in \Omega} \|P_t(x, \cdot) - \mu\|_{\text{TV}}.$$

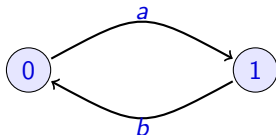
- ε -mixing time: for $\varepsilon \in (0, 1)$,

$$t_{\text{mix}}(\varepsilon) := \inf \{t \geq 0 : d(t) \leq \varepsilon\}, \quad t_{\text{mix}} := t_{\text{mix}}(1/4).$$

d is non-increasing, and sub-multiplicative: $d(s + t) \leq 2d(s)d(t)$.

A chain we can solve by hand: two states

$\Omega = \{0, 1\}$, rates $r(0, 1) = a > 0$, $r(1, 0) = b > 0$.



- Equilibrium: $\mu = \left(\frac{b}{a+b}, \frac{a}{a+b} \right)$.
- Solving the 2×2 linear system: $P_t(0, 1) = \frac{a}{a+b} (1 - e^{-(a+b)t})$, hence

$$d(t) = \frac{\max(a, b)}{a + b} e^{-(a+b)t}, \quad t_{\text{mix}}(\varepsilon) = \frac{1}{a + b} \log \frac{\max(a, b)}{(a + b)\varepsilon}.$$

- Eigenvalues of $\mathcal{L}$: 0 and $-(a + b)$.

Exponential decay at the rate given by the smallest nonzero eigenvalue.
Keep this in mind.

Spectral gap and relaxation time

Assume irreducibility and reversibility.

- **Spectral gap** = smallest nonzero eigenvalue of $-\mathcal{L}$:

$$\text{gap} := \inf_{f: \text{Var}_\mu(f) > 0} \frac{-\langle f, \mathcal{L}f \rangle_\mu}{\text{Var}_\mu(f)}, \quad \text{Var}_\mu(f) := \langle f, f \rangle_\mu - \langle f, \mathbf{1} \rangle_\mu^2.$$

- **Relaxation time**: $t_{\text{rel}} := 1/\text{gap}$.

Comparison

With $\mu_{\min} := \min_{x \in \Omega} \mu(x)$, for $\varepsilon \in (0, 1)$:

$$t_{\text{rel}} \log \frac{1}{2\varepsilon} \leq t_{\text{mix}}(\varepsilon) \leq t_{\text{rel}} \log \frac{1}{2\varepsilon \mu_{\min}}, \quad \lim_{t \rightarrow \infty} \frac{1}{t} \log d(t) = -\text{gap}.$$

The gap between the two bounds is a factor $\log(1/\mu_{\min}) \approx \log |\Omega|$ — **usually huge**. Which one is the truth?

Sequences of chains

A single finite chain always mixes; the question is only interesting asymptotically.

Consider a sequence $(\Omega_n, \mathcal{L}_n, \mu_n)_{n \in \mathbb{N}}$ with $|\Omega_n| \rightarrow \infty$, and write $t_{\text{mix}}^{(n)}(\varepsilon)$, $d_n(t)$.

Q: how does $t_{\text{mix}}^{(n)}(\varepsilon)$ grow with n , and how does it depend on ε ?

- Hypercube (n coordinates, each resampled at rate 1): $t_{\text{mix}}^{(n)} \approx \frac{1}{2} \log n$, $t_{\text{rel}} = 1$.
- Random walk on the cycle $\mathbb{Z}_n$: $t_{\text{mix}}^{(n)} \asymp n^2$, $t_{\text{rel}} \asymp n^2$.

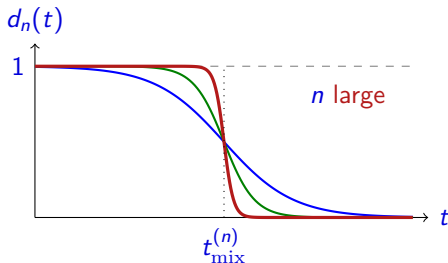
In the first case $t_{\text{rel}} \ll t_{\text{mix}}$; in the second they are of the same order. This difference is exactly the cutoff phenomenon.

The cutoff phenomenon

Definition (Aldous–Diaconis, 1986)

The sequence exhibits **cutoff** if for all $\varepsilon \in (0, 1)$,

$$\lim_{n \rightarrow \infty} \frac{t_{\text{mix}}^{(n)}(\varepsilon)}{t_{\text{mix}}^{(n)}(1 - \varepsilon)} = 1 \quad \Longleftrightarrow \quad \lim_{n \rightarrow \infty} d_n \left(c t_{\text{mix}}^{(n)} \right) = \begin{cases} 1 & \text{if } c < 1, \\ 0 & \text{if } c > 1. \end{cases}$$



The profile becomes a **step function**: the chain is far from equilibrium, and then suddenly it is not.

Cutoff on the hypercube: the proof in one slide

$\Omega_n = \{0, 1\}^n$; each coordinate has an independent Poisson clock of rate 1 and is resampled by a fair coin when it rings. Start from $x = (0, \dots, 0)$.

- **Lower bound.** Count $W(x) = \sum_{i \leq n} x_i$. A coordinate is still un-refreshed with probability e^{-t} , so

$$\mathbb{E}^x[W_t] = \frac{n}{2} - \frac{n e^{-t}}{2}, \quad \text{Var}(W_t) \leq \frac{n}{4}, \quad \mathbb{E}^\mu[W] = \frac{n}{2}, \quad \text{Var}_\mu(W) = \frac{n}{4}.$$

The bias $\frac{n e^{-t}}{2}$ beats the noise $\sqrt{n}$ as long as $t \leq \frac{1}{2} \log n - c$: Chebyshev gives $d_n(t) \rightarrow 1$.

- **Upper bound.** The chain is a product of n two-state chains, so a Fourier (L^2) computation gives

$$4 d_n(t)^2 \leq (1 + e^{-2t})^n - 1 \xrightarrow[t = \frac{1}{2}(\log n + c)]{} e^{e^{-c}} - 1.$$

Cutoff at time $\frac{1}{2} \log n$, with a window of size $O(1)$.

Back to the cards

Riffle shuffle (Bayer–Diaconis, 1992)

For a deck of n cards the riffle shuffle has cutoff at $\frac{3}{2} \log_2 n$ shuffles. For $n = 52$: about 7 shuffles.

Random transpositions (Diaconis–Shahshahani, 1981)

Pick two cards uniformly and exchange them: cutoff at $\frac{1}{2} n \log n$.

Top-to-random (Aldous–Diaconis, 1986)

Move the top card to a uniform position: cutoff at $n \log n$, by a coupon-collector argument.

Common feature: a sharp threshold, and a constant that can be identified. Cutoff was discovered precisely in this context.

When does cutoff hold?

Product condition

$$\text{gap}_n \cdot t_{\text{mix}}^{(n)} \longrightarrow \infty \quad \Longleftrightarrow \quad t_{\text{rel}}^{(n)} \ll t_{\text{mix}}^{(n)}.$$

- Cutoff $\implies$ product condition (for reversible chains).
- The converse is **false** in general: there are counterexamples (see Levin–Peres, Ch. 18).
- It is **true** in several natural classes: birth-and-death chains (Ding–Lubetzky–Peres), trees, random walks on some families of graphs ...
- **Diaconis' question (1996): find a general criterion.** Still open.

Consequence: proving cutoff for a given model is usually a **model-by-model** affair, and requires sharp bounds on both sides.

The standard toolbox

Upper bounds (the chain is close to μ):

- **Coupling.** Run two copies $(X_t), (Y_t)$ with $Y_0 \sim \mu$, each marginally correct. Then

$$d(t) \leq \mathbb{P}[X_t \neq Y_t].$$

- Strong stationary times; L^2 / spectral bounds; censoring (Peres–Winkler); path/log-Sobolev arguments.

Lower bounds (the chain is still far from μ):

- Find a **distinguishing statistic** $f : \Omega \rightarrow \mathbb{R}$ with

$$\mathbb{E}^x[f(X_t)] \text{ far from } \mathbb{E}^\mu[f], \quad \text{fluctuations of } f \text{ under both small.}$$

Chebyshev then separates the two laws.

- **Wilson's method** (2004): take f = eigenfunction of $\mathcal{L}$ with eigenvalue $-\lambda$; then $\mathbb{E}^x[f(X_t)] = e^{-\lambda t} f(x)$ decays at a known rate.

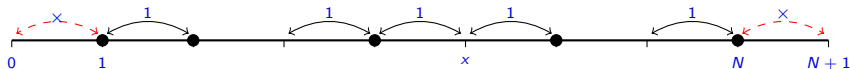
Part 3 puts these two ideas to work.

Part 3: The simple exclusion process

in dimension one

The simple exclusion process (SEP)

k indistinguishable particles on the segment $\llbracket 1, N \rrbracket$.



- (A) Each site holds at most one particle (the exclusion rule); the number of particles is conserved.
 - (B) On each edge $\{x, x+1\}$, $1 \leq x < N$, place an independent Poisson clock of rate 1. When it rings, swap the contents of the two sites.
- (No clock on $\{0, 1\}$ and $\{N, N+1\}$: the particles stay in the segment.)

Setup

- State space (1: particle, 0: empty site):

$$\Omega_{N,k} := \left\{ \xi : \llbracket 1, N \rrbracket \rightarrow \{0, 1\} \mid \sum_{x=1}^N \xi(x) = k \right\}, \quad |\Omega_{N,k}| = \binom{N}{k}.$$

- Generator: for $f : \Omega_{N,k} \rightarrow \mathbb{R}$,

$$(\mathcal{L}_{N,k} f)(\xi) := \sum_{x=1}^{N-1} [f(\xi \circ \tau_{x,x+1}) - f(\xi)],$$

where $\tau_{x,x+1}$ is the transposition of x and $x+1$.

- The uniform measure $\mu_{N,k}$ on $\Omega_{N,k}$ satisfies detailed balance,

$$\mu_{N,k}(\xi) \mathcal{L}_{N,k}(\xi, \eta) = \mu_{N,k}(\eta) \mathcal{L}_{N,k}(\eta, \xi),$$

hence it is the (unique) invariant probability measure.

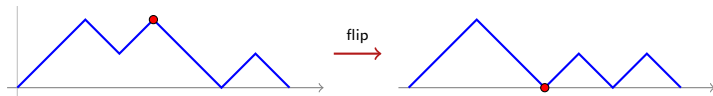
Question: starting from the worst configuration, how long until the particles are “uniformly placed”?

Three faces of the same chain

- $k = 1$: a single particle = random walk on $\llbracket 1, N \rrbracket$.
- **Labelled particles** ($k = N$, all distinct): the **interchange process**, i.e. the adjacent transposition shuffle — shuffling a deck by exchanging neighbouring cards.
- **Height function** $h : \llbracket 0, N \rrbracket \rightarrow \mathbb{Z}$,

$$h(0) = 0, \quad h(x) = \sum_{y=1}^x (2\xi(y) - 1).$$

SEP becomes the **corner-flip dynamics** of a lattice path.



A particle/hole swap = a local maximum turning into a local minimum.

Step 1: the spectral gap

One particle ($k = 1$): the generator is the discrete Laplacian (Neumann b.c.). Everything is explicit:

$$g_i(x) = \cos\left(\frac{i\pi(x - 1/2)}{N}\right), \quad \mathcal{L}_{N,1} g_i = -\lambda_i g_i, \quad \lambda_i = 2(1 - \cos(i\pi/N)),$$
$$\text{gap}_{N,1} = 2(1 - \cos(\pi/N)) = (1 + o(1))\frac{\pi^2}{N^2}.$$

Aldous' spectral gap conjecture (Caputo–Liggett–Richthammer, JAMS 2010)

For the interchange process on any finite weighted graph, the spectral gap equals that of the [single-particle](#) random walk. On the segment:

$$\text{gap}_{N,k} = \text{gap}_{N,1} = (1 + o(1))\frac{\pi^2}{N^2}, \quad \forall k \in \llbracket 1, N-1 \rrbracket.$$

So $t_{\text{rel}} \approx N^2/\pi^2$, independently of the number of particles.

Step 2: the mixing time

The general comparison only gives

$$\frac{N^2}{\pi^2} \log \frac{1}{2\varepsilon} \lesssim t_{\text{mix}}^{N,k}(\varepsilon) \lesssim \frac{N^2}{\pi^2} \cdot N.$$

Theorem (Lacoin, Ann. Probab. 2016)

If $\min(k, N - k) \rightarrow \infty$, then for every $\varepsilon \in (0, 1)$,

$$t_{\text{mix}}^{N,k}(\varepsilon) = (1 + o(1)) \frac{N^2 \log k}{2\pi^2},$$

so the simple exclusion process on the segment exhibits **cutoff**.

- The truth is the lower bound **times $\log k$** , not times N .
- Earlier: $t_{\text{mix}} \asymp N^2 \log N$ (Wilson 2004); general graphs (Oliveira 2013).
- Same statement for the **adjacent transposition shuffle** (labelled particles).

Where does $\frac{N^2 \log k}{2\pi^2}$ come from?

A two-line heuristic, via the height function h_t .

(i) **Averages follow the heat equation.** The generator acts on h as the discrete Laplacian, so

$$\partial_t \mathbb{E}[h_t] = \Delta \mathbb{E}[h_t] \implies \text{first Fourier mode decays like } e^{-\pi^2 t / N^2}.$$

(ii) **Worst start.** All k particles packed on the left: the initial height profile has amplitude of order k .

(iii) **Equilibrium fluctuations.** Under $\mu_{N,k}$ the height function fluctuates on the scale $\sqrt{k}$.

Mixing occurs when the deterministic profile shrinks below the equilibrium noise:

$$k e^{-\pi^2 t / N^2} \approx \sqrt{k} \iff t \approx \frac{N^2 \log k}{2\pi^2}.$$

The factor $\frac{1}{2}$ is exactly “signal k versus noise $\sqrt{k}$ ”.

Making the lower bound rigorous

Test function built from the first eigenfunction of the one-particle walk:

$$\Phi(\xi) := \sum_{x=1}^N g_1(x) \xi(x), \quad g_1(x) = \cos\left(\frac{\pi(x - 1/2)}{N}\right).$$

- Exact decay: since $\mathcal{L}_{N,1} g_1 = -\text{gap } g_1$, the particle density evolves linearly and

$$\mathbb{E}^\xi [\Phi(\xi_t)] = e^{-\text{gap } t} \Phi(\xi).$$

Starting with the particles packed on the left, $\Phi(\xi_0) \asymp k$.

- Second moment: $\text{Var}(\Phi) = O(k)$, both along the dynamics and under $\mu_{N,k}$ (negative correlations help).
- Chebyshev: as long as $e^{-\text{gap } t} k \gg \sqrt{k}$, the events $\{\Phi > a\}$ separate P_t^ξ from $\mu_{N,k}$, so $d_{N,k}(t) \approx 1$.

$$\implies t_{\text{mix}}^{N,k}(\varepsilon) \geq (1 - o(1)) \frac{N^2 \log k}{2\pi^2}. \quad (\text{This is Wilson's method.})$$

Making the upper bound rigorous

Harder: one must show the chain **is** mixed, not just that a statistic is.

- **Monotone coupling**. The height functions are partially ordered and the dynamics preserves the order. It is enough to couple the two **extremal** initial conditions $h^{\max}$ and $h^{\min}$:

$$d_{N,k}(t) \leq \mathbb{P} [h_t^{\max} \neq h_t^{\min}] .$$

- **Area argument** (Wilson). The area $A_t = \sum_x (h_t^{\max}(x) - h_t^{\min}(x)) \geq 0$ is a supermartingale decaying at rate **gap**, so $\mathbb{E}[A_t] \leq e^{-\text{gap} t} A_0$ becomes **$o(1)$** slightly after $\frac{N^2 \log k}{2\pi^2}$.
- **Censoring** (Peres–Winkler). Removing updates can only make the chain less mixed — a licence to run the dynamics in a convenient order (e.g. on sub-segments first).

Coalescence then happens within $o(N^2 \log k)$ extra time.

What happens if we perturb the model?

- **Asymmetric jumps (ASEP)**: particles prefer one direction. Mixing time of order N instead of $N^2 \log N$, with cutoff (Labbé–Lacoin).
- **Inhomogeneous / random environment**: put a conductance $c(x, x+1) > 0$ on each edge, i.i.d. say.

$$\text{gap}_{N,k} = \text{gap}_{N,1}, \quad t_{\text{mix}}^{N,k}(\varepsilon) = (1 + o(1)) \frac{N^2 \log k}{2\pi^2}$$

under a law of large numbers for the resistances $1/c$ [Y. '24+]. The homogenized medium behaves like the homogeneous one — but heavy tails, traps and slow bonds change the answer.

- **Higher dimension**, general graphs, other conservative dynamics (zero-range process, ...): largely open.

A general principle: “one particle determines the whole system” — when it is true, and when it fails.

Summary

- A Markov chain on a finite state space converges to equilibrium; **the mixing time** says how fast.
- Spectral gap gives the asymptotic rate, but can be off by a factor $\log |\Omega|$; determining the constant requires real work.
- **Cutoff**: convergence happens abruptly. Proved case by case; a general criterion is an open problem.
- **Simple exclusion on $[1, N]$** : gap $= \pi^2/N^2$ (Aldous' conjecture), cutoff at $\frac{N^2 \log k}{2\pi^2}$ — heat equation + equilibrium fluctuations.

Suggested reading

- D. Levin, Y. Peres, *Markov Chains and Mixing Times*, 2nd ed., AMS 2017. (Chapters 1–6, 12, 18.)
- P. Diaconis, *The cutoff phenomenon in finite Markov chains*, PNAS 1996.
- T. Liggett, *Interacting Particle Systems*, Springer.

Thank you!

Obrigado! Xièxie!

Questions?